# Chapter 6

## Partial Differentiation

## Lecture content: Functions of Several variables:

Example 1(a) (page 888), Example 4, Example 6

Exercise: 14.1 (13,15,16,18,30,31)

**Sample Example 1:** Determine the domain of each of the followings

**(a)** $f(x,y)=\sqrt{x+y}$,

**(b)** $f(x,y)=\sqrt{x}+\sqrt{y}$,

**(c)** $f(x,y)=\ln\left(9-x^2-9y^2\right)$.

**Solution:(a)** In this case we know that we can't take the square root of a negative number so this means that we must require, $x+y\geq 0$.

Here is a sketch of the graph of this region.

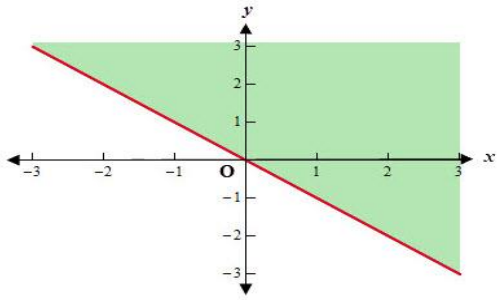



Fig. 1

(b) This function is different from the function in the previous part. Here we must require that,

$$x\geq 0, y\geq 0$$

and they really do need to be separate inequalities. There is one for each square root in the function. Here is the sketch of this region.

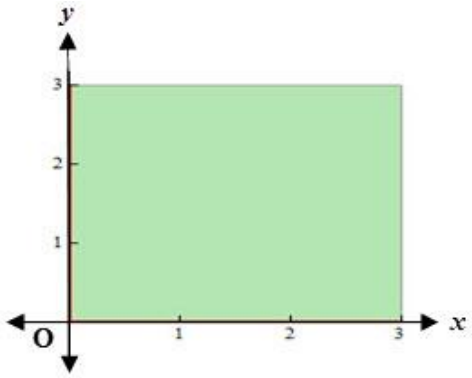



Fig. 2

**(c)** In this final part we know that we can't take the logarithm of a negative number or zero. Therefore we need to require that,

$$9-x^2-9y^2>0 \text{ or, } \frac{x^2}{9}+\frac{y^2}{1}<1$$

and upon rearranging we see that we need to stay interior to an ellipse for this function. Here is a sketch of this region.

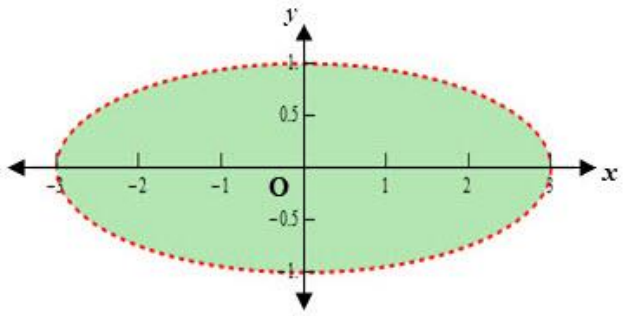



Fig. 3

## Sample Exercise:

**1.** Describe the region (domain) on which the following functions are continuous.

(a) $\frac{xy^2}{\sqrt{x^2+y^2-4}}$, (b) $\cos\left(\frac{x^2y^2}{16-x^2+y^2}\right)$, (c) $\sqrt{x-2y}$, (d) $\frac{x}{\sqrt{16-x^2-y^2}}$

(e) $\sqrt{1-\frac{x^2}{16}-\frac{y^2}{9}}$, (f) $\ln(x+y-3)$ and (g) $\frac{4}{x^2+y^2}$.

## Lecture content: Interpretation of Partial derivatives:

**Example 2, 4, 5,8,9,10 (page 915-920)**

## Exercise 14.3 41,42,43,63,64,67,68,76

**Example 2:** Find partial derivatives $f_x$, $f_y$, $f_{xx}$ and $f_{yy}$ of the function $f(x,y)=\sqrt{x^2+y^2}$ .

**Solution**: $\dfrac{\partial f}{\partial x}=f_x=\dfrac{1}{2\sqrt{x^2+y^2}}\times 2x=\dfrac{x}{\sqrt{x^2+y^2}}$, $\dfrac{\partial f}{\partial y}=f_y=\dfrac{1}{2\sqrt{x^2+y^2}}\times 2y=\dfrac{y}{\sqrt{x^2+y^2}}$

$$\frac{\partial^2 f}{\partial x^2}=f_{xx}=\frac{\sqrt{x^2+y^2}-x\times\dfrac{x}{\sqrt{x^2+y^2}}}{\left(x^2+y^2\right)}=\frac{y^2}{(x^2+y^2)^{3/2}}$$

$$\frac{\partial^2 f}{\partial y^2}=f_{yy}=\frac{\sqrt{x^2+y^2}-x\times\dfrac{x}{\sqrt{x^2+y^2}}}{\left(x^2+y^2\right)}=\frac{x^2}{\left(x^2+y^2\right)^{3/2}}$$

**Theorem:** If $f(x,y)$ is continuous and $f_{xy}$ and $f_{yx}$ exist and are continuous then $f_{xy}=f_{yx}$.

**Example 3**: Show that $f_{xy}=f_{yx}$ for the function $f(x,y)=\sqrt{x^2+y^2}$ .

**Solution:** $f(x,y)=\sqrt{x^2+y^2}$

$$\therefore \frac{\partial f}{\partial x}=f_x=\frac{1}{2\sqrt{x^2+y^2}}\times 2x=\frac{x}{\sqrt{x^2+y^2}}$$

$$\frac{\partial^2 f}{\partial x\partial y}=f_{xy}=\frac{\partial}{\partial y}\frac{x}{\sqrt{x^2+y^2}}=-\frac{xy}{(x^2+y^2)^{3/2}}$$

Similarly, $\dfrac{\partial^2 f}{\partial y\partial x}=f_{yx}=-\dfrac{xy}{(x^2+y^2)^{3/2}}$

Therefore, $f_{xy}=f_{yx}$.

## 6.2 Differential

The total differential (differential) of a function $w = f(x, y, z)$ is defined by the equation

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz$$

Whether or not $x, y \,\&\, z$ are independent of each other, provided only that the partial derivatives involves are continuous.

**Example 4**: Find the differential of the function $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$.

**Solution:** We have $R = \frac{R_1 R_2}{R_1 + R_2}, \frac{\partial R}{\partial R_1} = \frac{R_2^2}{(R_1 + R_2)^2}, \frac{\partial R}{\partial R_2} = \frac{R_1^2}{(R_1 + R_2)^2}$

Total differential is $dR = \frac{\partial R}{\partial R_1}dR_1 + \frac{\partial R}{\partial R_2}dR_2 = \frac{R_2^2}{(R_1 + R_2)^2}dR_1 + \frac{R_1^2}{(R_1 + R_2)^2}dR_2$

**Example 5**: The ideal gas law state that $pV = RT$, where p (pressure), V (volume) and T (temperature), R (=10 say) is a constant. Find approximate change of V when p decreases from 1 to 0.99 and T decreases from 5 to 4.99.

**Solution:** We have $V = \frac{10T}{p}$

Total differential of $V$ at (1, 5) is $dV = \frac{\partial V}{\partial p}dp + \frac{\partial V}{\partial T}dT = 10dT - 50dp$.

Here $dT = -0.01, dp = -0.01$

Hence approximate change in V of $dV = 10(-0.01) - 50(-0.01) \approx 0.4$.

## 6.3 Chain Rule

**Chain Rule I:** Suppose $u = f(x, y, z)$ has continuous partial derivatives and let $x = x(t)$ $y = y(t)$ and $z = z(t)$ be differentiable. Then we can consider $f$ as a function of single variable $t$, we have from above differential

$$\frac{df}{dt}=\frac{\partial f}{\partial x}\frac{dx}{dt}+\frac{\partial f}{\partial y}\frac{dy}{dt}+\frac{\partial f}{\partial z}\frac{dz}{dt}.$$

**Chain Rule II:** Suppose $u=f(x,y)$ has continuous partial derivatives and let $x=x(s,t)$ and $y=y(s,t)$ be differentiable with continuous partial derivatives. Then

$$\frac{\partial f}{\partial s}=\frac{\partial f}{\partial x}\frac{\partial x}{\partial s}+\frac{\partial f}{\partial y}\frac{\partial y}{\partial s}$$

$$\frac{\partial f}{\partial t}=\frac{\partial f}{\partial x}\frac{\partial x}{\partial t}+\frac{\partial f}{\partial y}\frac{\partial y}{\partial t}\ .$$

**Example 6:** If $V=f\left(x-y,y-z,z-x\right)$, prove that $\frac{\partial V}{\partial x}+\frac{\partial V}{\partial y}+\frac{\partial V}{\partial z}=0$.

**Solution:** Let $V=f\left(p,q,r\right)$ where $p=x-y,\quad q=y-z,\quad r=z-x$.

Now $\frac{\partial p}{\partial x}=1,\quad \frac{\partial p}{\partial y}=-1,\quad \frac{\partial p}{\partial z}=0$,

$$\frac{\partial q}{\partial x}=0,\quad \frac{\partial q}{\partial y}=1,\quad \frac{\partial q}{\partial z}=-1$$

and $\frac{\partial r}{\partial x}=-1,\quad \frac{\partial r}{\partial y}=0,\quad \frac{\partial r}{\partial z}=1$.

Using Chain rule , we can write

$$\begin{aligned}\frac{\partial V}{\partial x}&=\frac{\partial V}{\partial p}\frac{\partial p}{\partial x}+\frac{\partial V}{\partial q}\frac{\partial q}{\partial x}+\frac{\partial V}{\partial r}\frac{\partial r}{\partial x}\\&=\frac{\partial V}{\partial p}1+\frac{\partial V}{\partial q}0+\frac{\partial V}{\partial r}(-1)\\&=\frac{\partial V}{\partial p}-\frac{\partial V}{\partial r}\end{aligned}$$

Similarly, $\frac{\partial V}{\partial y}=-\frac{\partial V}{\partial p}+\frac{\partial V}{\partial q}$ and $\frac{\partial V}{\partial z}=-\frac{\partial V}{\partial q}+\frac{\partial V}{\partial r}$.

Therefore,

$$\frac{\partial V}{\partial x}+\frac{\partial V}{\partial y}+\frac{\partial V}{\partial z}$$
$$=\frac{\partial V}{\partial p}-\frac{\partial V}{\partial r}-\frac{\partial V}{\partial p}+\frac{\partial V}{\partial q}-\frac{\partial V}{\partial q}+\frac{\partial V}{\partial r}$$
$$=0.$$

## 6.4 Implicit Differentiation $\left(\frac{dy}{dx}\right)$ using Partial Derivatives

Suppose that the equation $F(x,y)=0$ has a solution $y=f(x)$, where $f(x)$ has continuous derivative. Then

$\frac{dy}{dx}=-\frac{\partial F/\partial x}{\partial F/\partial y}=-\frac{F_x}{F_y}$, where $F_y\neq 0$.

**Example 7:** Find $\frac{dy}{dx}$ using partial differentiation for $x^3y+xy^3=2$.

**Solution:** Let, $F(x,y)=x^3y+xy^3-2=0$

$$F_x=3x^2y+y^3$$

$$F_y=x^3+3xy^2$$

Therefore, $\frac{dy}{dx}=-\frac{3x^2y+y^3}{x^3+3xy^2}$.

**Partial differentiation for the system of equations is shown through examples:**

**Example 8:** If $2u+3v=\sin x$ and $u+2v=x\cos y$, find $u_x,u_y,v_x,v_y$.

**Solution:** $2u+3v=\sin x$

$u+2v=x\cos y$

Solving this two equations for $u$ and $v$ we get,

$$u=2\sin x-3x\cos y \text{ …….(1) and}$$

$$v=2x\cos y-\sin x \text{ ……..(2)}$$

Differentiating equation no (1) & (2) w.r.t. $x$, we get

$u_x=2\cos x-3\cos y$ and $v_x=2\cos y-\cos x$.

Differentiating equation no (1) & (2) w.r.t. $y$, we get

$u_y = 3x\sin y$ and $v_y = -2x\sin y$.

**Example 9:** Given $u^2 + x^2 + y^2 = 3$ and $u - v^3 + 3x = 4$, find $u_x, u_y, v_x, v_y$.

**Solution:** $u^2 + x^2 + y^2 = 3 \ldots\ldots(1)$

$u - v^3 + 3x = 4 \ldots\ldots\ldots(2)$

Differentiating equation no (1) & (2) w.r.t. $x$, we get

$2uu_x + 2x = 0$ and $u_x - 3v^2 v_x + 3 = 0$

Solving this two equations for $u_x$ and $v_x$ by Crammer's rule we get

$$u_x = \frac{\begin{vmatrix} -2x & 0 \\ -3 & -3v^2 \end{vmatrix}}{\begin{vmatrix} 2u & 0 \\ 1 & -3v^2 \end{vmatrix}} = \frac{6xv^2}{-6uv^2} = -\frac{x}{u} \quad \text{and } v_x = \frac{\begin{vmatrix} 2u & -2x \\ 1 & -3 \end{vmatrix}}{\begin{vmatrix} 2u & 0 \\ 1 & -3v^2 \end{vmatrix}} = \frac{-6u + 2x}{-6uv^2} = \frac{3u - x}{3uv^2}.$$

Differentiating equation no (1) & (2) w.r.t. $y$, we get

$2uu_y + 2y = 0$ and $u_y - 3v^2 v_y = 0$

Solving this two equations for $u_y$ and $v_y$ by Crammer's rule we get

$$u_y = \frac{\begin{vmatrix} -2y & 0 \\ 0 & -3v^2 \end{vmatrix}}{\begin{vmatrix} 2u & 0 \\ 1 & -3v^2 \end{vmatrix}} = \frac{6yv^2}{-6uv^2} = -\frac{y}{u} \quad \text{and } v_y = \frac{\begin{vmatrix} 2u & -2y \\ 1 & 0 \end{vmatrix}}{\begin{vmatrix} 2u & 0 \\ 1 & -3v^2 \end{vmatrix}} = \frac{2y}{-6uv^2} = -\frac{y}{3uv^2}.$$

### 6.5 Homogeneous Function

A function $f(x, y)$ is said to homogeneous of degree *n* if

$$f(kx, ky) = k^n f(x, y)$$

where *k* is a constant.

**Alternative:** A function $f(x, y)$ is said to homogeneous of degree *n* if

$$f(x, y) = x^n f\left(\frac{y}{x}\right) \text{ or, } f(x, y) = y^n f\left(\frac{x}{y}\right)$$

A similar definition applies to a function of any number of independent variables. For example,

$$f(x,y,z)=x^2+yz+z^2$$

is a homogeneous function of degree 2 in *x*, *y* and *z* since

$$f(kx,ky,kz)=(kx)^2+ky(kz)+(kz)^2=k^2\left(x^2+yz+z^2\right),$$

Or,

$$f(x,y,z)=x^2\left(1+\frac{y}{x}\times\frac{z}{x}+\frac{z^2}{x^2}\right) \text{ or, } f(x,y,z)=y^2\left(\frac{x^2}{y^2}+\frac{z}{y}+\frac{z^2}{y^2}\right), \text{ or,}$$

$$f(x,y,z)=z^2\left(\frac{x^2}{z^2}+\frac{y}{z}+1\right)$$

**6.6 Euler's Theorem:** If $f(x,y)$ is a homogeneous function of two variables *x* and *y* of degree *n*, then $x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=nf$.

**Example 10:** Test whether the function $f=\frac{x^2+y^2}{x+y}$ is homogeneous or not. If homogeneous verify Euler's theorem for this function.

**Solution:** Here,

$$f(\lambda x,\lambda y)=\frac{(\lambda x)^2+(\lambda y)^2}{\lambda x+\lambda y}=\lambda\left(\frac{x^2+y^2}{x+y}\right)=\lambda\, f(x,y)$$

Therefore $f(x,y)$ is a homogeneous function of order 1.

Applying Euler's theorem to this function we have $x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=f$.

$$f(x,y)=\frac{x^2+y^2}{x+y}$$

$$\Rightarrow \frac{\partial f}{\partial x}=\frac{(x+y)2x-\left(x^2+y^2\right)}{(x+y)^2}=\frac{x^2-y^2+2xy}{(x+y)^2}.$$

Similarly, $\frac{\partial f}{\partial y}=\frac{(x+y)2y-\left(x^2+y^2\right)}{(x+y)^2}=\frac{y^2-x^2+2xy}{(x+y)^2}$.

and

$$x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=x\frac{x^2-y^2+2xy}{(x+y)^2}+y\frac{y^2-x^2+2xy}{(x+y)^2}$$

$$=x\frac{x^2-y^2+2xy}{(x+y)^2}+y\frac{y^2-x^2+2xy}{(x+y)^2}$$

$$=\frac{(x+y)(x^2+y^2)}{(x+y)^2}==\frac{(x^2+y^2)}{(x+y)}=f(x,y).$$

Hence the Euler's theorem is verified.

**Application of Euler's theorem**

**Example11:** If $u=\operatorname{arc}\csc\sqrt[3]{\frac{x^2+y^2+z^2}{x^3+y^3+z^3}}$, find $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}$.

**Solution:** Let $f(x,y,z)=\csc u=\sqrt[3]{\frac{x^2+y^2+z^2}{x^3+y^3+z^3}}$.

Here $f(x,y,z)=\sqrt[3]{\frac{x^2+y^2+z^2}{x^3+y^3+z^3}}$ is a homogeneous function of degree $-\frac{1}{3}$.

According to Euler's theorem,

$$x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}+z\frac{\partial f}{\partial z}=-\frac{1}{3}f(x,y,z)$$

$$\Rightarrow\quad x\frac{\partial}{\partial x}(\csc u)+y\frac{\partial}{\partial y}(\csc u)+z\frac{\partial}{\partial z}(\csc u)=-\frac{1}{3}\csc y$$

$$\Rightarrow\quad x\frac{\partial}{\partial u}(\csc u)\frac{\partial u}{\partial x}+y\frac{\partial}{\partial u}(\csc u)\frac{\partial u}{\partial y}+z\frac{\partial}{\partial u}(\csc u)\frac{\partial u}{\partial z}=-\frac{1}{3}\csc u$$

$$\Rightarrow\quad x(-\csc u\cot u)\frac{\partial u}{\partial x}+y(-\csc u\cot u)\frac{\partial u}{\partial y}+z(-\csc u\cot u)\frac{\partial u}{\partial z}=-\frac{1}{3}\csc u$$

$$\Rightarrow\quad -\csc u\cot u\left(x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}\right)=-\frac{1}{3}\csc u$$

$$\Rightarrow\quad x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}=\frac{1}{3}\tan u.$$

## 6.7 Relative Extrema and Their Nature

**Lecture Content: page 959 : Example 1,Example 2**

**Page: 961 Second derivative Test: P-961, Example 3, 4,5, 6**

**Exercise:14.7 5,6,7**

We shall now consider the extreme values for functions of two variables.

Suppose $f(x,y)$ and all its derivatives up to order two are continuous and $f_x(a,b) = f_y(a,b) = 0$. Then

(a) $(a,b)$ is a local maximum if $f_{xx}(a,b) < 0$ or $f_{yy}(a,b) < 0$ and $D > 0$.

(b) $(a,b)$ is a local minimum if $f_{xx}(a,b) > 0$ or $f_{yy}(a,b) > 0$ and $D > 0$.

(c) $(a,b)$ is a saddle point if $D < 0$.

(d) no conclusion can be drawn if $D = 0$.

where $D = f_{xx} f_{yy} - (f_{xy})^2$ evaluated at $(a,b)$.

**Example12:** Find the critical points of $f(x,y) = 4 + x^3 + y^3 - 3xy$ .Also state their nature.

**Solution:** We first need all the first order (to find the critical points) and second order (to classify the critical points) partial derivatives so let's get those.

$f_x = 3x^2 - 3y,\ \ f_y = 3y^2 - 3x,$
$f_{xx} = 6x,\qquad f_{yy} = 6y$

Now, Critical points will be solutions to the system of equations,

$f_x = 0, f_y = 0,$

$\Rightarrow 3x^2 - 3y = 0, 3y^2 - 3x = 0,$

Solving $(x,y) = (0,0),(1,1)$.

**Nature:** $D(x,y) = 36xy - 9$ **,** $D(0,0) = 9 < 0$

So, for $(0,0)$ , $D$ is negative and so this must be a saddle point.

$D(1,1) = 27 > 0$ and $f_{xx}(1,1) = 6 > 0$

So, for $(1,1)$, $D$ is positive and $f_{xx}$ is positive and so we must have a relative minimum.

The following figures show the graph of *f(x,y)*.

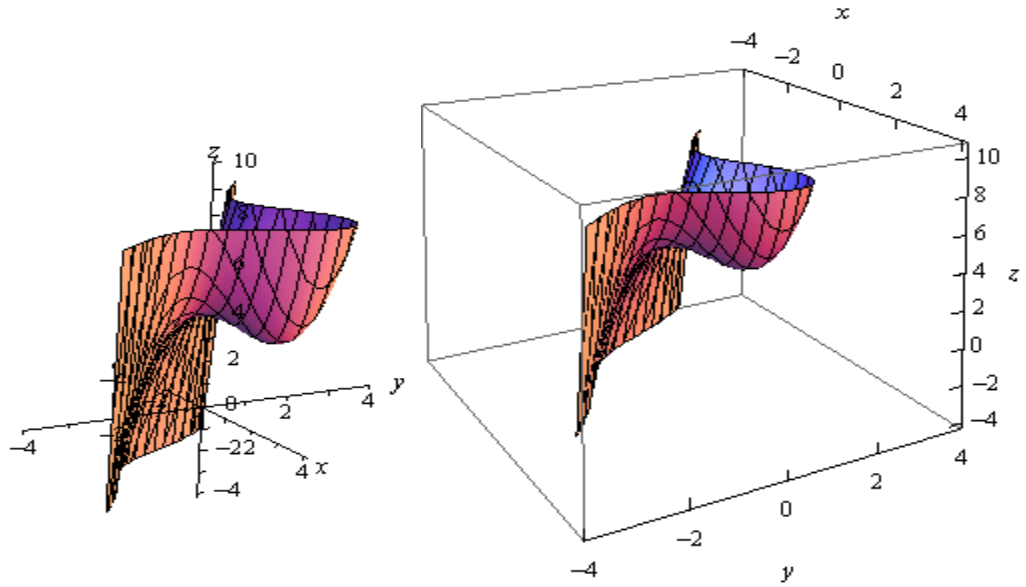



**Fig 4:** *f(x, y)*

**Example 13:** Determine the point on the plane $4x - 2y + z = 1$ that is closest to the point $(-2, -1, 5)$.

**Solution:** Let's suppose that $(x, y, z)$ is any point on the plane. The distance between this point and the point in question, $(-2, -1, 5)$ is given by the formula,

$$d = \sqrt{(x+2)^2 + (y+1)^2 + (z-5)^2}$$

Given that, $4x - 2y + z = 1$ or $z = 1 - 4x + 2y$

Substituting this into the distance formula gives,

$$d = \sqrt{(x+2)^2 + (y+1)^2 + (1-4x+2y-5)^2}$$
$$= \sqrt{(x+2)^2 + (y+1)^2 + (-4-4x+2y)^2}$$

Now, notice that finding the minimum value of *d* will be equivalent to finding the minimum value of $d^2$.

So,

$$f(x,y) = d^2 = (x+2)^2 + (y+1)^2 + (-4-4x+2y)^2$$

Now,

$$f_x = 2(x+2)+2(-4)(-4-4x+2y) = 36+34x-16y$$
$$f_y = 2(y+1)+2(2)(-4-4x+2y) = -14-16x+10y$$
$$f_{xx} = 34$$
$$f_{yy} = 10$$
$$f_{xy} = -16$$

and,

$$D = 34(10)-(-16)^2 = 84 > 0$$

so any critical points that we get will be relative minimums.
Now let's find the critical point(s).

$$36+34x-16y = 0$$
$$-14-16x+10y = 0$$

Solving these two equations we get we get a single critical point: $\left(-\frac{34}{21}, -\frac{25}{21}\right)$.
So,

$$z = 1-4\left(-\frac{34}{21}\right)+2\left(-\frac{25}{21}\right) = \frac{107}{21}$$

So, the point on the plane that is closest to $(-2,-1,5)$ is $\left(-\frac{34}{21}, -\frac{25}{21}, \frac{107}{21}\right)$.

**6.8Lagrange Multipliers**

The problem of finding extrema subject to side condition called constraint may be obtained by eliminating one of the variable using constraint equation.

For example to find extrema of $f(x,y,z)$ subject to $g(x,y,z)=0$, we may proceed as follows.

Define a new function

$$H(x,y,\lambda) = f(x,y) - \lambda g(x,y)$$

and considering the equations

$$H_x = \frac{\partial H}{\partial x} = 0, \qquad H_y = \frac{\partial H}{\partial y} = 0, \qquad H_\lambda = \frac{\partial H}{\partial \lambda} = 0.$$

To find the extrema of $f(x,y,z)$ subject to the constraints $g_1(x,y,z)=0$ and $g_2(x,y,z)=0$, define a function with two Lagrange multipliers $\lambda_1$ and $\lambda_2$ as

$$H(x,y,z,\lambda_1,\lambda_2) = f(x,y,z) - \lambda_1 g_1(x,y,z) - \lambda_2 g_2(x,y,z)$$

and then solve the equations

$$H_x = H_y = H_z = 0,\ H_{\lambda_1} = g_1(x,y,z) = 0 \text{ and } H_{\lambda_2} = g_2(x,y,z) = 0$$

**Example 14:** Find the points on $x^2 + y^2/4 + z^2 = 33$ at which $x - 2y + 4z$ attains its maximum and minimum.

**Solution:** Let, $H(x,y,z,\lambda) = x - 2y + 4z - \lambda\left(x^2 + \frac{y^2}{4} + z^2 - 33\right)$

The equations for the conditional extrema gives

$$H_x = 1 - 2\lambda x = 0 \qquad \text{or} \qquad x = \frac{1}{2\lambda}$$

$$H_y = -2 - \frac{\lambda y}{2} = 0 \qquad \text{or} \qquad y = -\frac{4}{\lambda}$$

$$H_z = 4 - 2\lambda z = 0 \qquad \text{or} \qquad z = \frac{2}{z}$$

Substituting those values in the equation

$$H_\lambda = -\left(x^2 + \frac{y^2}{4} + z^2 - 33\right) = 0$$

we have $$\frac{1}{4\lambda^2} + \frac{4}{\lambda^2} + \frac{4}{\lambda^2} = 33$$

$$132\lambda^2 = 33 \qquad \text{or} \qquad \lambda = \pm\frac{1}{2}$$

The maxima and minima then occur at $(1,-8,4)$ and $(-1,8,-4)$.

Since $f(x,y,z) = x - 2y + 4z$,

$$f(1,-8,4) = 1 + 16 + 16 = 33$$

and $$f(-1,8,-4) = -1 - 16 - 16 = -33$$

Thus $f(x,y,z) = x - 2y + 4z$ has a maximum at $(1,-8,4)$, and a minimum at $(-1,8,-4)$.

## Sample Exercise 6.1

**2.** Find $f_{xx}$, $f_{xy}$, $f_{yx}$, $f_{yy}$, $f_{xxx}$, and $f_{yyy}$ where

(a) $f(x,y) = 3xy^2 + xy$, (b) $f(x,y) = \exp(2xy) + \frac{x}{y}$,

(c) $f(x,y) = (x^3 + y\text{^}2)\sin(y - x)$ and (d) $f(x,y) = \exp(\cosh x) + \sin(y^2 - x^2)$.

**3.** Find $f_{xz}$, $f_{yz}$, $f_{zx}$, $f_{zy}$, $f_{zz}$ , $f_{xyz}$, $f_{xzy}$, $f_{yzx}$ and $f_{zzz}$ where

(a) $f(x,y,z) = \exp(y)\ln x + x^2y^2 - zx^3$ and (b) $f(x,y,z) = x^3yz^2 + xz - yz^2$

**4.** Evaluate the indicated partial derivatives.

(a) $w = y^2\cos(xy)$, $\frac{\partial w}{\partial x}\left(\pi, \frac{1}{2}\right)$ and $\frac{\partial w}{\partial y}\left(\pi, \frac{1}{2}\right)$.

(b) $z = \sqrt{x^3 + 2\,y^2}$, $\frac{\partial z}{\partial x}(2,1)$and$\frac{\partial z}{\partial y}(2,1)$.

**5.** Verify the **Laplace equation** $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, where

(a) $u = x^2 - y^2 + 2xy$, (b) $u = \exp(2y)\sin(2x)$ and (c) $u = \exp(y)\sin x + \exp(x)\sin y$

**6.** Show that the following functions satisfy one dimensional **heat equation**

$$\frac{\partial z}{\partial t} = c^2\frac{\partial^2 z}{\partial x^2} \quad (c > 0\text{, constant})$$

(a) $z = \exp(-t)\sin\left(\frac{x}{c}\right)$ and (b) $z = \exp(-t)\cos\left(\frac{x}{c}\right)$.

**7.** (i) Show that $u = x^3 - 3xy^2$, and $v = 3x^2y - y^3$ satisfies **Cauchy-Riemann equations** in rectangular form $u_x = v_y$ and $u_y = -v_x$.

(ii) Show that $u = r^4\cos 4\theta$, $v = r^4\sin 4\theta$, satisfies **Cauchy-Riemann equations** in polar form $u_r = \frac{1}{r}v_\theta$ and $v_r = -\frac{1}{r}u_\theta$.

**8.** Define homogeneous function. Test whether the following functions are homogeneous. If homogeneous, what is the degree of these functions?

(a) $f(x,y,z) = yz + xy + zx$, (b) $f(x,y) = \sqrt{x + yx}$, (c) $f(x,y) = \frac{x^2 - y^2}{\sqrt[5]{x} + \sqrt[5]{y}}$

**9.** State **Euler's theorem**.Verify **Euler's theorem** for the following functions:

(a) $f(x,y,z) = x^2 - y^2 - z^2$, (b) $f(x,y) = \frac{xy}{y - x}$,

(c) $f(x,y) = \sqrt{x^5 + y^5}$, (d) $f(x,y,z) = x^2\,z^2 + y^2xz$

**10.** (a) Show that the function $f(x,y) = \exp\left(\frac{y}{x}\right)$ satisfies the relation $xf_x + yf_y = 0$.

(b) Prove the relation in part (a) using **Euler's theorem**.

**11.** If $u = \arctan\left(\frac{x^3 + y^3 + z^3}{x - y - z}\right)$, prove that $x\,u_x + y\,u_y + z\,u_z = \sin 2u$.

**12.** If $u = \operatorname{arccsc}\left(\frac{x^2 + y^2 + z^2}{\sqrt{x} - \sqrt{y} - \sqrt{z}}\right)$, prove that $x\,u_x + y\,u_y + z\,u_z = -\frac{3}{2}\tan u$.

**13.** Use partial derivatives to find $\frac{dy}{dx}$, given that

(a) $x^2 - 5y^3 = 6$, (b) $y - \exp(y)\cos(xy)$, (c) $xy - y^2 + 3xy - 9 = 0$, (d) $x^y + y^{2x} = 6$

and (e) $e^{xy} + x^2 e^x = 2$. **Also check your result using implicit differentiation.**

**14.** Suppose that$w = \exp(xyz)$, $x = 3u + v$, $y = 2u + v$ and $z = uv^2$. Use appropriate forms of the **chain rule** to evaluate $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$.

**15.** Given that $T = x^3y - x^2 y^3 + 5$, $x = r\cos\theta$, $y = r\sin\theta$. Use **chain rule** to find $\frac{\partial T}{\partial r}$ and $\frac{\partial T}{\partial \theta}$.

**16.** Given that $z = \sqrt{xy + x}$, $x = \cos\theta$, $y = \sin\theta$. Use **chain rule** to find $\frac{dz}{d\theta}$ when $\theta = \frac{\pi}{2}$.

**17.** If $u = y^2\sin x + x^2\cos y$, $x = \cos 2t$, $y = \sin 2t$ then use **chain rule** to find $\frac{du}{dt}$.

## Sample Exercise 6.2

1. Find the total differential of the following functions:
   (a) $f(x, y, z) = \exp(x^2)\cos(y + z)$, (b) $g(x, y) = \sin(y - z^2)$ and
   (c) $h(x, y, z) = y^2x^{-3}z^{-\frac{3}{2}}$
2. The altitude of a right circular cone is 12 inches and is increasing at 0.4 in/min. the radius of the base is 10 inches and is decreasing at 0.2 in/min. How fast is the volume changing?
3. Examine the following functions for maxima, minima and saddle points:
   (a) $f(x,y) = x^3 - 3xy + y^3$,
   (b) $f(x,y) = 12x + 6y - x^2 + xy - y^2$.
4. Find all the relative maxima, relative minima, and saddle points (if any) of the following functions:
   (a) $z = x^2y^2 - x^2 - y^2$, (b) $z = 2x^3 - 6xy + 3y^2$,
   (c) $z = xy - x^3 - y^2$, (d) $z = x^3 + y^3 + 3x^2 - 3y^2 - 8$, (e) $z = x^4 + 4x^2y^2 - 2x^2 + 2y^2 - 1$.